An induction proof of the secant-number contraction identity

A deterministic polynomial calculation

Let r, a be independent polynomial variables, and define the linear operator on $\mathbb{R}[r, a]$

$$H = 4r\partial_r^2 + 4\partial_r + r\partial_a^2. \tag{1}$$

Write $E_m = \frac{d^m}{dz^m} \sec z \Big|_{z=0}$ for the secant numbers: $E_0 = 1$, $E_2 = 1$, $E_4 = 5$, $E_6 = 61$, and $E_m = 0$ for odd m. We prove $(H^m a^m)(0, 0) = 2^m m! E_m$ through a stronger identity that is closed under induction.

For $n, k, \ell \in \mathbb{N}_0$, set

$$d_{n,k} = \frac{d^n}{dz^n}(\sec z \, \tan^k z)\Big|_{z=0}, \qquad T_{k,\ell} = (H^{k+\ell}(r^k a^\ell))(0, 0). \tag{2}$$

Here H^0 is the identity operator, and all derivatives in z are taken in a neighbourhood of zero.

Theorem 1 (Mixed-monomial contraction)**.** *For every $k, \ell \in \mathbb{N}_0$,*

$$\boxed{T_{k,\ell} = 2^{2k+\ell} k! \, \ell! \, d_{k+\ell,k}.} \tag{3}$$

Consequently, for every $m \in \mathbb{N}_0$,

$$\boxed{(H^m a^m)(0, 0) = 2^m m! E_m.} \tag{4}$$

The two recurrences. Direct differentiation gives

$$H(r^k a^\ell) = 4k^2 r^{k-1} a^\ell + \ell(\ell - 1) r^{k+1} a^{\ell-2}. \tag{5}$$

Throughout, the first term is omitted when $k = 0$, and the second when $\ell < 2$; negative powers and negative-index quantities are never used. For $N = k + \ell \geq 1$, applying H^{N-1} and evaluating at zero yields

$$T_{k,\ell} = 4k^2 T_{k-1,\ell} + \ell(\ell - 1) T_{k+1,\ell-2}, \tag{6}$$

with the same omitted-term convention. Each term present has total index $N - 1$.

Also,

$$\frac{d}{dz}(\sec z \, \tan^k z) = k \sec z \, \tan^{k-1} z + (k + 1) \sec z \, \tan^{k+1} z,$$

where the first term is omitted for $k = 0$. Hence, for $n \geq 0$,

$$d_{n+1,k} = k \, d_{n,k-1} + (k + 1) d_{n,k+1}. \tag{7}$$

Finally, the Taylor expansion of $\sec z \, \tan^k z$ has leading term z^k, so

$$d_{n,k} = 0 \qquad \text{whenever } n < k. \tag{8}$$

Induction proof

Proof of Theorem 1. We use strong induction on $N = k + \ell$. For $N = 0$, necessarily $k = \ell = 0$, and $T_{0,0} = 1 = d_{0,0}$, establishing (3).

Let $N \geq 1$ and assume (3) for every pair whose total index is less than N. Fix $k, \ell \geq 0$ with $k + \ell = N$, and abbreviate

$$C_{k,\ell} = 2^{2k+\ell} k!\, \ell!.$$

If $k \geq 1$, the induction hypothesis gives

$$\begin{aligned} 4k^2 T_{k-1,\ell} &= 4k^2\, 2^{2k+\ell-2}(k-1)!\, \ell!\, d_{N-1,k-1} \\ &= C_{k,\ell}\, k\, d_{N-1,k-1}. \end{aligned}$$

If $\ell \geq 2$, it similarly gives

$$\begin{aligned} \ell(\ell-1) T_{k+1,\ell-2} &= \ell(\ell-1)\, 2^{2k+\ell}(k+1)!(\ell-2)!\, d_{N-1,k+1} \\ &= C_{k,\ell}\, (k+1) d_{N-1,k+1}. \end{aligned}$$

When $\ell = 0$ or 1, the second term in (6) is absent. The corresponding term required by the recurrence for d vanishes, since

$$N - 1 = k + \ell - 1 < k + 1 \quad \Longrightarrow \quad d_{N-1,k+1} = 0$$

by (8). When $k = 0$, the first term is absent in both recurrences.

Substituting into (6) and then using (7), we obtain

$$T_{k,\ell} = C_{k,\ell}\big(k\, d_{N-1,k-1} + (k+1) d_{N-1,k+1}\big) = C_{k,\ell} d_{N,k},$$

again omitting the first term if $k = 0$. This completes the induction. Taking $k = 0$, $\ell = m$, and using $d_{m,0} = E_m$ proves (4). □

Relation to the horizontal differential operator

Consider the operator on polynomials in (x, y, a),

$$\widetilde{H} = (\partial_x - y\partial_a)^2 + (\partial_y + x\partial_a)^2.$$

Expanding it gives

$$\widetilde{H} = \partial_x^2 + \partial_y^2 + 2(x\partial_y - y\partial_x)\partial_a + (x^2 + y^2)\partial_a^2.$$

For $p \in \mathbb{R}[r, a]$ and $r = x^2 + y^2$, the angular term vanishes, while $(\partial_x^2 + \partial_y^2) p(r, a) = 4r p_{rr} + 4p_r$. Therefore

$$\widetilde{H}\,[p(x^2 + y^2, a)] = (Hp)(x^2 + y^2, a).$$

Iterating this identity and evaluating at the origin gives the equivalent form of the contraction formula:

$$\boxed{(\widetilde{H}^{\,m} a^m)(0, 0, 0) = 2^m m! E_m.}$$

Scope. This is a deterministic proof of the polynomial contraction identity. The separate reduction from tensor-matrix coefficients to this operator is not proved here. The general induction has not yet been formalized in Lean.